Notes

Chapter 1

1. Lord Kelvin was quoted by the physicist Albert Michelson during his 1894 address at the dedication of the University of Chicago's Ryerson Laboratory (see D. Kleppner, *Physics Today,* November 1998).

2. Lord Kelvin, "Nineteenth Century Clouds over the Dynamical Theory of Heat and Light," *Phil. Mag.* Ii—6th series, 1 (1901).

3. A. Einstein, N. Rosen, and B. Podolsky, *Phys. Rev.* 47, 777 (1935).

4. Sir Arthur Eddington, *The Nature of the Physical World* (Cambridge, Eng.: Cambridge University Press, 1928).

5. As described more fully in note 2 of Chapter 6, this is an overstatement because there are examples, involving relatively esoteric particles (such as K-mesons and B-mesons), which show that the so-called weak nuclear force does not treat past and future fully symmetrically. However, in my view and that of many others who have thought about it, since these particles play essentially no role in determining the properties of everyday material objects, they are unlikely to be important in explaining the puzzle of time's arrow (although, I hasten to add, no one knows this for sure). Thus, while it is technically an overstatement, I will assume throughout that the error made in asserting that the laws treat past and future on equal footing is minimal—at least as far as explaining the puzzle of time's arrow is concerned.

6. Timothy Ferris, *Coming of Age in the Milky Way* (New York: Anchor, 1989).

Chapter 2

1. Isaac Newton, *Sir Isaac Newton's Mathematical Principle of Natural Philosophy and His System of the World,* trans. A. Motte and